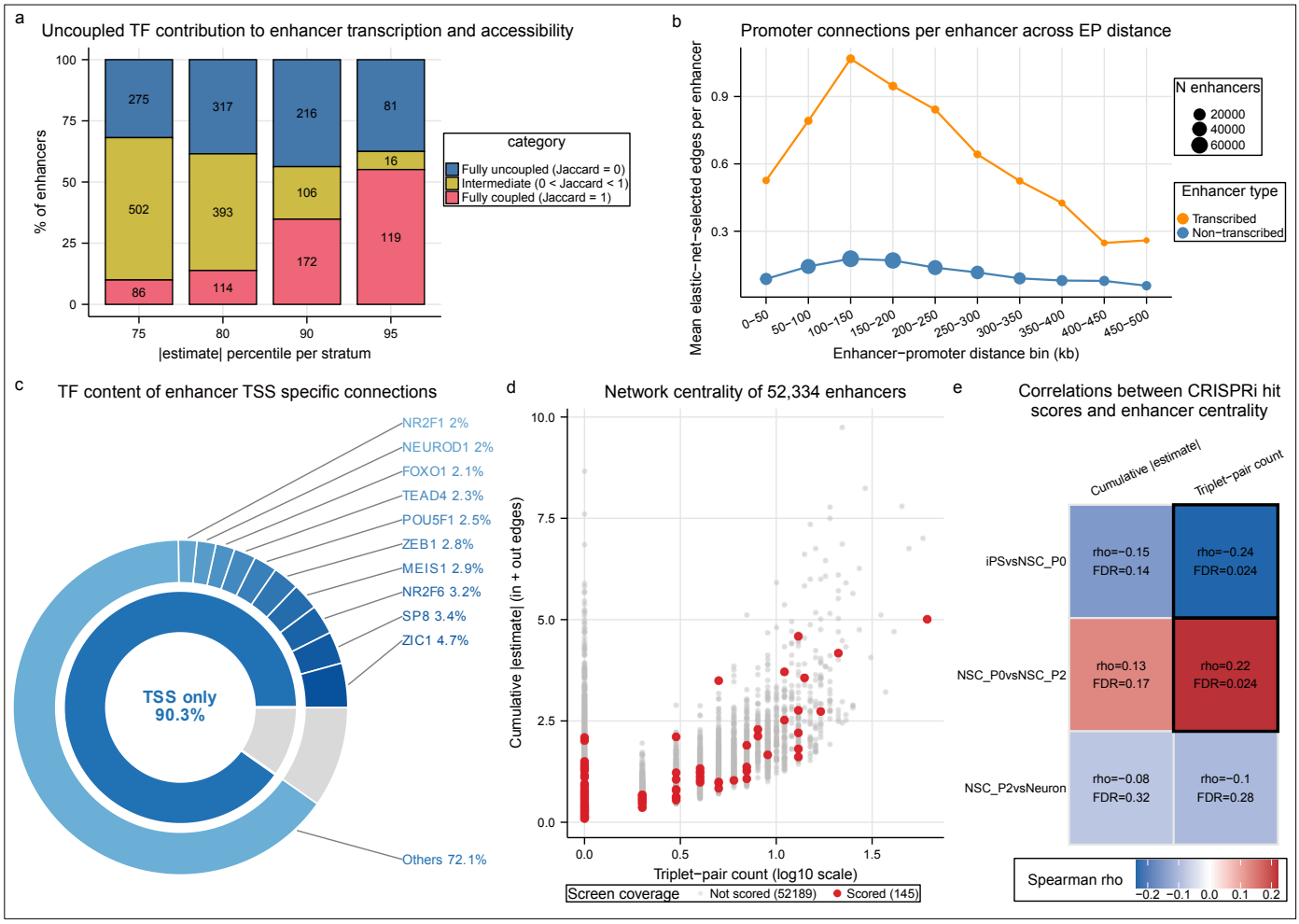



**Figure 6 | Enhancer transcription reveals distinct regulatory interactions and network hubs.**

**a**, Overlap between TF sets associated with enhancer accessibility and transcription across four thresholds. Each bar represents one percentile threshold of absolute edge coefficients, with the 90th-percentile threshold corresponds to the final high-confidence GRN. Colors indicate whether accessibility- and transcription-associated TF sets are completely distinct, partially overlapping or completely overlapping. **b**, Mean number of Lasso-selected enhancer–promoter edges per enhancer across 50-kb genomic-distance bins up to 500 kb, stratified by transcribed (orange) and non-transcribed (blue) enhancers. Dot size represents the number of enhancers in each distance bin. **c**, Regulatory routes uniquely recovered by incorporating enhancer transcription into the bimodal GRN. The inner ring partitions 3,929 TF–enhancer–promoter triplets absent from the accessibility-only network into new routes, for which both TF–enhancer and enhancer–promoter edges were novel (dark blue, 90.3%), and routes sharing at least one edge with the accessibility-only network (grey, 9.7%). The outer ring shows the contribution of the TFs associated with the new routes. Incorporating enhancer transcription added 3,098 TF–enhancer and 3,390 enhancer–promoter edges. **d**, Network centrality of 52,334 enhancers quantified by the number of unique TF–promoter pairs bridged through each enhancer (x-axis) and the cumulative absolute elastic-net coefficient of connected edges (y-axis). Red points indicate 145 enhancers targeted in the CRISPRi neurogenesis screen with available hit scores. **e**, Spearman correlations between CRISPRi hit scores and enhancer centrality across three differentiation contrasts for the 145 screen-scored enhancers. Color represents Spearman's rho (red, positive; blue, negative); black outlines indicate FDR < 0.05.